ARTIFICIAL INTELLIGENCE (CT 653)

Tribhuvan University | Institute of Engineering | BCT | Year III / Part II

CHAPTER 6: Machine Learning

Syllabus Hours: 6 hrs | Marks Weight: 10 marks

Complete Exam-Ready Answers | All Past Questions Covered

CHAPTER 6 SYLLABUS TOPICS

- 6.1 Concepts of Learning
- 6.2 Learning by Analogy, Inductive Learning, Explanation Based Learning
- 6.3 Neural Networks
- 6.4 Genetic Algorithm
- 6.5 Fuzzy Learning
- 6.6 Boltzmann Machines

Prepared by: Santosh Kushwaha (ACE079BCT)

SECTION 1: Concepts of Learning — Types of Machine Learning

Syllabus Ref: 6.1, 6.2 | Exam Ref: 2082 Kartik, 2081 Ashwin, 2080 Ashwin Back, 2079 Jestha, 2078 Poush, 2074 Bhadra, 2073 Bhadra

Q1. Define Machine Learning. What is Fuzzy logic? Explain Fuzzy inference with suitable example. [1+2+5] (2080 Ashwin Back)

Answer: Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to LEARN from experience (data) and improve their performance on a task WITHOUT being explicitly programmed for every case.

Formal Definition (Tom Mitchell, 1997): A computer program is said to LEARN from experience E with respect to some task T and performance measure P, if its performance at task T, as measured by P, improves with experience E.

Example: An email spam filter learns from labeled emails (E), improves its spam detection (T), and is measured by accuracy (P).

Types of Machine Learning:

Type	Description	How it Learns	Examples
Supervised Learning	Learns from labeled training data (input-output pairs)	Minimizes error between predicted and actual output	Classification, Regression, Decision Trees, Neural Nets
Unsupervised Learning	Learns from unlabeled data — finds hidden patterns/structure	Groups similar data together (clustering) or reduces dimensions	K-Means Clustering, PCA, Autoencoders
Reinforcement Learning	Learns by interacting with environment and receiving rewards/penalties	Maximizes cumulative reward through trial and error	Game playing (Chess, Go), Robot navigation
Semi-Supervised Learning	Uses small amount of labeled + large amount of unlabeled data	Combines supervised and unsupervised techniques	Text classification with few labeled examples

Q2. What is Machine Learning? Discuss about Supervised, Unsupervised and Reinforcement Learning with appropriate examples. [2+6] (2078 Poush)

Answer: Supervised vs Unsupervised vs Reinforcement Learning

1. Supervised Learning

In Supervised Learning, the algorithm is trained on a LABELED dataset — each input has a corresponding correct output (label). The algorithm learns a mapping function f(x) → y.

Process:

1. Collect labeled training data: {(x1,y1), (x2,y2), ..., (xn,yn)}

2. Train the model — adjust parameters to minimize prediction error.
3. Test the model on unseen data to evaluate performance.
4. Use the model for predictions.

Types: Classification (output is a category) and Regression (output is a number).

Examples:

- Email spam detection: Input = email text, Output = Spam/Not Spam.
- House price prediction: Input = house features, Output = Price (number).
- Medical diagnosis: Input = symptoms, Output = Disease name.
- Algorithms: Decision Tree, SVM, Linear Regression, Neural Networks.

2. Unsupervised Learning

In Unsupervised Learning, the algorithm is trained on UNLABELED data. It must find hidden structure, patterns, or groupings on its own without any guidance.

Types: Clustering (grouping similar data) and Dimensionality Reduction (reducing features while preserving information).

Examples:

- Customer segmentation: Group customers by purchasing behavior (no predefined groups).
- Anomaly detection: Find unusual patterns in network traffic (no labeled attacks).
- Document clustering: Group similar news articles together automatically.
- Algorithms: K-Means, DBSCAN, PCA, Autoencoders, SOM (Self-Organizing Maps).

3. Reinforcement Learning

In Reinforcement Learning, an AGENT learns to make decisions by interacting with an ENVIRONMENT. The agent receives REWARDS for good actions and PENALTIES for bad actions. The goal is to maximize cumulative reward over time.

Key Components: Agent, Environment, State (S), Action (A), Reward (R), Policy (π).

Process: Agent observes state → takes action → receives reward → updates policy → repeats.

Examples:

- Game playing: AlphaGo learns to play Go by playing millions of games against itself.
- Robot navigation: Robot learns to reach a destination by trial and error.
- Recommendation systems: Learn which content to recommend based on user engagement.

Comparison Table:

Feature	Supervised	Unsupervised	Reinforcement
Data	Labeled (input + output)	Unlabeled (input only)	No static dataset — environment
Goal	Learn a mapping f(x)→y	Find hidden structure	Maximize cumulative reward
Feedback	Direct (error signal)	None (self-organized)	Delayed rewards/penalties
Output	Class label or number	Clusters or components	Policy (what action to take)
Example	Spam detection	Customer clustering	Game playing (Chess)
Key Algorithm	Decision Tree, SVM	K-Means, PCA	Q-Learning, Policy Gradient

Q3. What is machine learning? Explain learning by analogy with example. [2076 Baisakh, 2073 Bhadra, 2071 Bhadra] [2+6] (2076 Baisakh)

Answer: Learning by Analogy

Learning by Analogy (Case-Based Reasoning / Analogical Reasoning) is a type of machine learning where the system solves a NEW problem by finding a SIMILAR problem it has already solved, and adapting the old solution to the new situation.

Key Idea: If problem A is similar to problem B, and we know the solution to B, then we can adapt B's solution to solve A.

Steps in Learning by Analogy:

1. RETRIEVE: Find the most similar past case from the case base.
2. REUSE: Adapt the solution of the retrieved case to the new problem.
3. REVISE: Test and revise the adapted solution if necessary.
4. RETAIN: Save the new case (problem + solution) in the case base for future use.

This is also called Case-Based Reasoning (CBR).

Example 1 — Medical Diagnosis:

- New patient: 35-year-old male, fever 39°C, cough, breathing difficulty, no rash.
- Retrieved case: 32-year-old male, fever 38.5°C, cough, breathing difficulty, diagnosed with Pneumonia.
- Similarity: High (same symptoms, similar age/gender).
- Adapted solution: Diagnose new patient with Pneumonia, prescribe antibiotics.
- After confirmation, store new case for future reference.

Example 2 — Legal Reasoning:

- New case: Company A polluted a river and caused fish deaths.
- Analogous case: Company B polluted a lake in 1998 and was fined $2 million.
- Analogical reasoning: Company A should also be fined a similar amount.

Example 3 — Engineering Design:

- Design a bridge in a new location → Retrieve design of a similar bridge in similar terrain → Adapt the design for new conditions.

Q4. 'Learning is an essential characteristic for intelligent agents.' Comment on this statement. Differentiate between Supervised and Unsupervised Learning. [2079 Jestha] [4+4] (2079 Jestha)

Answer: Why Learning is Essential for Intelligent Agents

The statement 'Learning is an essential characteristic for intelligent agents' is fundamentally correct. Here is why:

1. ADAPTATION: The real world is dynamic and changing. An agent that cannot learn will fail when the environment changes. A learning agent can adapt to new situations.
2. INCOMPLETE KNOWLEDGE: Designers cannot anticipate every situation an agent will face. Learning allows the agent to acquire knowledge that wasn't explicitly programmed.
3. IMPROVEMENT: A learning agent improves its performance over time through experience, becoming more accurate and efficient.

4. GENERALIZATION: A learning agent can handle new cases by generalizing from past experience, not just repeating memorized responses.

5. AUTONOMY: Learning enables agents to operate autonomously without constant human guidance or reprogramming.

Examples: A chess-playing AI that learns from thousands of games becomes better than one with only preprogrammed rules. A spam filter that learns from user feedback improves over time.

Supervised vs Unsupervised Learning:

Feature	Supervised Learning	Unsupervised Learning
Training Data	Labeled (input-output pairs given)	Unlabeled (only inputs given)
Goal	Learn a function f(x)→y for prediction	Discover hidden patterns/structure
Human Involvement	High — labels must be provided	Low — no labels needed
Output Type	Specific label or numerical value	Cluster assignment or components
Evaluation	Easy — compare predicted vs actual label	Hard — no ground truth to compare
Use Case	When labeled data is available	When labeling is expensive/impossible
Examples	Image classification, spam detection	Customer segmentation, anomaly detection
Key Algorithms	Decision Tree, SVM, Neural Networks	K-Means, PCA, Autoencoders

Q5. Define inductive learning. Explain in detail about ID3 process with suitable example. [2+6] (2074 Bhadra)

Answer: Inductive Learning

Inductive Learning is a type of machine learning where the system learns general rules/patterns from SPECIFIC examples. It 'induces' (generalizes) from particular observations to a general hypothesis.

Example: After seeing 100 examples where 'fever + cough = flu', the system induces the rule: IF fever AND cough THEN flu.

ID3 Algorithm (Iterative Dichotomiser 3):

ID3 is a decision tree learning algorithm developed by J. Ross Quinlan. It builds a decision tree from a training dataset by selecting the best attribute at each node using INFORMATION GAIN.

Key Concept — Entropy: Entropy measures the impurity/disorder in a dataset.

```
Entropy(S) = -Σ p_i × log2(p_i)
```

Where p_i is the proportion of class i in the dataset S.

Information Gain:

```
Gain(S, A) = Entropy(S) - Σ (|Sv|/|S|) × Entropy(Sv)
```

Where Sv is the subset of S for which attribute A has value v.

Steps of ID3 Algorithm:

1. If all examples have the same class, return a leaf node with that class.
2. If no attributes remain, return a leaf node with the majority class.
3. Calculate Information Gain for each attribute.
4. Select the attribute with the HIGHEST Information Gain as the root/next node.
5. Split the dataset on the selected attribute into subsets.
6. Recursively apply steps 1-5 to each subset.

Example: Play Cricket Decision (from 2079 Jestha exam)

Given dataset: Outlook, Temperature, Humidity, Windy → Play Cricket (Yes/No)

Outlook	Temperature	Humidity	Windy	Play Cricket
Rainy	Hot	High	False	Yes
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	True	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes

(Total: 14 examples — Yes:9, No:5 — similar dataset from 2079 Jestha)

Step 1 — Calculate Overall Entropy:

```
S = 14 examples, Yes=9, No=5
Entropy(S) = -(9/14)log2(9/14) - (5/14)log2(5/14)
           = -(0.643 × (-0.637)) - (0.357 × (-1.485))
           = 0.410 + 0.530 = 0.940 bits
```

Step 2 — Calculate Gain for Outlook:

```
Sunny subset (5): Yes=3, No=2 → Entropy = 0.971
Overcast subset (4): Yes=4, No=0 → Entropy = 0 (pure!)
Rainy subset (5): Yes=2, No=3 → Entropy = 0.971
Gain(S, Outlook) = 0.940 - (5/14×0.971 + 4/14×0 + 5/14×0.971)
                 = 0.940 - (0.347 + 0 + 0.347) = 0.246 bits
```

Similarly calculate for other attributes. The attribute with HIGHEST gain is selected as root.

Result: Outlook has Gain=0.246, which is typically the highest, making it the root node. Overcast branch → always Yes (pure leaf). Sunny and Rainy branches are further split.

Decision Tree Structure:

```
              [OUTLOOK]
          /      |      \
      Sunny  Overcast  Rainy
        |       |        |
   [HUMIDITY][Yes]   [WINDY]
    /    \            /    \
```

```
High  Normal    True  False
 |       |       |      |
No     Yes      No    Yes
```

SECTION 2: Genetic Algorithm — Most Repeated in Chapter 6

Exam Ref: 2082 Kartik, 2081 Chaitra, 2080 Chaitra, 2079 Chaitra, 2079 Jestha, 2078 Chaitra, 2077 Chaitra, 2076 Baisakh, 2075 Baisakh, 2075 Bhadra, 2074 Bhadra, 2072 Ashwin

Q6. What is fuzzy logic? Explain the steps involved in Genetic Algorithm with examples. [2082 Kartik, 2081 Chaitra, 2078 Chaitra, 2077 Chaitra, 2075 Baisakh] [2+6] (Multiple years)

Answer: Genetic Algorithm (GA)

A Genetic Algorithm is a search and optimization technique inspired by the process of NATURAL SELECTION and BIOLOGICAL EVOLUTION (Darwin's theory). It is used to find optimal or near-optimal solutions to complex problems.

Biological Inspiration: Just as nature evolves better organisms over generations by selection, crossover, and mutation — GA evolves better solutions over generations.

Key Terminology:

GA Term	Biological Equivalent	Description
Individual / Chromosome	Organism	A single candidate solution
Gene	Gene	A single element/parameter of the solution
Population	Population	A set of candidate solutions
Fitness Function	Natural fitness / survival	Function that measures quality of a solution
Selection	Natural selection	Choose better-fit individuals to reproduce
Crossover (Recombination)	Sexual reproduction	Combine parts of two parents to create offspring
Mutation	Random genetic mutation	Randomly alter a gene to introduce diversity
Generation	Generation	One iteration of the GA loop

Steps of Genetic Algorithm:

1. INITIALIZATION: Generate an initial population of N random chromosomes (candidate solutions). Each chromosome is encoded (typically as a binary string or real-valued vector).

Example: Optimize $f(x) = x^2$. Population of 4: [1010, 0110, 1101, 0100] (binary, range 0-15).

2. FITNESS EVALUATION: Calculate the fitness of each individual using the fitness function. Higher fitness = better solution.

Chromosome	Decimal Value	Fitness $f(x) = x^2$
1010	10	100
0110	6	36
1101	13	169
0100	4	16

3. SELECTION: Select individuals for reproduction. Individuals with higher fitness have a higher probability of being selected. Common methods:

- Roulette Wheel Selection: Probability proportional to fitness.
- Tournament Selection: Randomly pick k individuals, choose the best.
- Rank Selection: Rank by fitness, select proportional to rank.

Example (Roulette): Total fitness = 100+36+169+16 = 321. P(1101) = 169/321 = 52.6% chance of selection.

4. CROSSOVER (Recombination): Two selected parent chromosomes exchange genetic material at a randomly chosen crossover point to produce two offspring.

Single-point crossover example:

```
Parent 1: 1010 | 1101
Parent 2: 0110 | 0100
Crossover at position 2:
Offspring 1: 1010 | 0100  = 10100100
Offspring 2: 0110 | 1101  = 01101101
```

Crossover Rate (Pc): Typically 0.6–0.9. Controls how often crossover occurs.

5. MUTATION: Randomly flip one or more bits (genes) in the offspring chromosome. This introduces new genetic material and prevents premature convergence.

```
Before mutation: 10100100
Flip bit at position 5: 10100000  (bit flipped 1→0)
```

Mutation Rate (Pm): Typically 0.001–0.01. Very low to avoid destroying good solutions.

6. REPLACEMENT: Replace the old population with the new population of offspring. Sometimes keep the best individual (elitism).

7. TERMINATION: Repeat steps 2-6 for a specified number of generations OR until a satisfactory fitness level is achieved.

Full GA Flow (Block Diagram):

```
Start
  |
  v
[Initialize Population]
  |
  v
[Evaluate Fitness of each individual]
  |
  v
[Termination condition met?] --- Yes ---> [Return Best Solution] ---> Stop
  |
  No
  |
  v
[Selection — choose parents based on fitness]
  |
  v
[Crossover — create offspring]
  |
  v
[Mutation — randomly alter genes]
  |
  v
[Replace population with new offspring]
```

```
|
+-----> [Evaluate Fitness] (loop back)
```

GA Operators Summary:

Operator	Purpose	Analogy	Common Types
Selection	Choose fit individuals for reproduction	Survival of the fittest	Roulette Wheel, Tournament, Rank
Crossover	Combine two parents to produce offspring	Sexual reproduction	Single-point, Two-point, Uniform
Mutation	Randomly alter a gene	Random genetic mutation	Bit-flip, Swap, Inversion

Applications of Genetic Algorithm:

- Traveling Salesman Problem (TSP) — finding the shortest route.
- Job scheduling and resource allocation.
- Neural network training — optimizing weights.
- Game playing and strategy optimization.
- Engineering design — structural and circuit optimization.

Q7. What is evolutionary computing? List down the steps in genetic algorithm with an example. What do you understand by swarm intelligence? [2079 Chaitra, 2075 Bhadra] [2+6] (2079 Chaitra)

Answer: Evolutionary Computing & Swarm Intelligence

Evolutionary Computing: Evolutionary Computing is a family of optimization algorithms inspired by biological evolution. It includes:

Technique	Inspired By	Key Feature
Genetic Algorithms (GA)	Genetic evolution (Darwin)	Binary/real-coded chromosome, crossover, mutation
Genetic Programming (GP)	Evolution of programs	Evolves computer programs as tree structures
Evolution Strategies (ES)	Self-adaptive mutation	Real-valued parameters, self-adapting step sizes
Differential Evolution (DE)	Population-based optimization	Vector differences for generating trial solutions

Swarm Intelligence: Swarm Intelligence is a collective behavior of decentralized, self-organized systems — inspired by the social behavior of insects and animals (ants, bees, birds, fish).

Algorithm	Inspired By	Application
Ant Colony Optimization (ACO)	Ants finding shortest path using pheromones	Routing, TSP
Particle Swarm Optimization (PSO)	Birds flocking / fish schooling	Continuous optimization

Bee Algorithm	Honey bee foraging behavior	Function optimization
Firefly Algorithm	Firefly bioluminescence	Multimodal optimization

The GA steps are the same as described in Q6 above.

SECTION 3: Fuzzy Learning / Fuzzy Logic

Exam Ref: 2082 Kartik, 2080 Chaitra, 2079 Ashwin Back, 2078 Baishakh, 2077 Chaitra, 2076 Baisakh, 2074 Bhadra, 2073 Magh

Q8. What is fuzzy logic? When shall we use it? Explain the steps involved in Fuzzy Inference with suitable example. [1+2+5] (2079 Ashwin Back, 2080 Chaitra)

Answer: Fuzzy Logic

Fuzzy Logic is a form of many-valued logic that deals with APPROXIMATE REASONING rather than precise, binary (true/false) reasoning. Introduced by Lotfi Zadeh (1965), it allows truth values to be ANY real number between 0 and 1 (instead of just 0 or 1).

Classical Logic: A person is either TALL (1) or NOT TALL (0). Binary.

Fuzzy Logic: A person of 5'8" might be TALL with membership degree 0.7 — partially tall.

Key Concepts:

Concept	Description	Example
Fuzzy Set	A set where elements have degrees of membership (0 to 1)	TALL = {5'6"→0.3, 5'8"→0.6, 6'0"→1.0}
Membership Function	Function that maps input to membership degree [0,1]	μ_tall(x) = (x-150)/50 for x in [150,200]
Linguistic Variable	A variable with fuzzy values like 'cold', 'warm', 'hot'	Temperature: {Cold, Warm, Hot}
Fuzzy Rule	IF-THEN rule using linguistic variables	IF temp is HOT THEN fan speed is HIGH
Defuzzification	Converting fuzzy output back to a crisp value	Centroid: find center of area of fuzzy output

When to Use Fuzzy Logic:

- When the problem involves imprecise, vague, or subjective information.
- When mathematical model is too complex or unknown.
- When approximate reasoning is acceptable and preferable to rigid rules.
- Examples: Washing machine control, air conditioner, camera autofocus, car braking, medical diagnosis.

Steps in Fuzzy Inference System (Mamdani Method):

1. FUZZIFICATION: Convert crisp (precise) input values to fuzzy values using membership functions.

2. RULE EVALUATION: Apply fuzzy IF-THEN rules to determine the degree to which each rule fires.

3. AGGREGATION: Combine the outputs of all rules into a single fuzzy set.

4. DEFUZZIFICATION: Convert the aggregated fuzzy output back to a crisp value.

Complete Example: Air Conditioner Controller

Problem: Control fan speed based on room temperature.

Linguistic Variable (Temperature): COLD (< 15°C), WARM (15–25°C), HOT (> 25°C)

Linguistic Variable (Fan Speed): SLOW, MEDIUM, FAST

Fuzzy Rules:

- Rule 1: IF temperature is COLD THEN fan speed is SLOW
- Rule 2: IF temperature is WARM THEN fan speed is MEDIUM
- Rule 3: IF temperature is HOT THEN fan speed is FAST

Input: Temperature = 28°C (a crisp value)

Step 1 — Fuzzification:

```
μ_COLD(28) = 0.0  (28°C is not cold at all)
μ_WARM(28) = 0.2  (28°C is slightly warm)
μ_HOT(28)  = 0.8  (28°C is mostly hot)
```

Step 2 — Rule Evaluation:

```
Rule 1 fires with strength: 0.0 → SLOW at 0.0
Rule 2 fires with strength: 0.2 → MEDIUM at 0.2
Rule 3 fires with strength: 0.8 → FAST at 0.8
```

Step 3 — Aggregation: Combine the truncated fuzzy sets for SLOW, MEDIUM, FAST into one output fuzzy set.

Step 4 — Defuzzification (Centroid Method): Find the center of mass of the aggregated fuzzy output area.

```
Defuzzified Fan Speed = Σ(μ(x) × x) / Σ(μ(x))
≈ 75% of max speed (i.e., fan runs at high speed)
```

Conclusion: For 28°C input, the fuzzy controller outputs a fan speed of approximately 75%, which is between MEDIUM and FAST — a smooth, realistic control action.

SECTION 4: Boltzmann Machines

Syllabus Ref: 6.6 | Exam Ref: 2075 Bhadra, 2072 Ashwin, 2070 Magh, 2076 Baisakh

Q9. What is machine learning? Explain in detail about Boltzmann Machines with suitable algorithm and explanations. [2+8] (2072 Ashwin)

Answer: Boltzmann Machine

A Boltzmann Machine is a type of STOCHASTIC (probabilistic) RECURRENT NEURAL NETWORK developed by Geoffrey Hinton and Terrence Sejnowski (1985). It is a generative model — it learns the probability distribution of training data and can generate new samples from that distribution.

Named After: Ludwig Boltzmann (physicist who developed statistical mechanics and the energy concept used in the model).

Structure:

Component	Description
Visible Units (v)	Represent observed data (inputs). These are the data nodes.
Hidden Units (h)	Represent latent/hidden factors. These capture patterns in data.
Connections (Weights)	Every unit is connected to every other unit (fully connected). No self-connections.
Bias	Each unit has a bias term that influences its activation threshold.
Stochastic Activation	Each unit is turned ON (1) or OFF (0) with a probability based on its energy.

Energy Function: The Boltzmann Machine uses an energy function to define the probability of each state:

```
E(v, h) = -Σ(i,j) w_ij × v_i × h_j - Σ_i b_i × v_i - Σ_j c_j × h_j
```

Where w_ij = weight between visible unit i and hidden unit j, b_i = bias of visible unit i, c_j = bias of hidden unit j.

Probability of a State:

```
P(v, h) = (1/Z) × exp(-E(v, h))
Where Z = partition function (normalizing constant) = Σ exp(-E(v,h))
```

States with LOWER energy have HIGHER probability of occurring.

Types of Boltzmann Machines:

Type	Structure	Key Feature
Full Boltzmann Machine	All units connected to all other units	Theoretically powerful but computationally expensive
Restricted Boltzmann Machine (RBM)	No intra-layer connections (visible-visible or hidden-hidden)	Efficient training using Contrastive Divergence (CD)
Deep Belief Network (DBN)	Stack of RBMs	Enables deep learning with greedy layer-wise training

Training Algorithm: Contrastive Divergence (CD-1)

Used for Restricted Boltzmann Machines (most practical form):

1. POSITIVE PHASE (Reality): Clamp visible units to training data. Compute the probability of hidden units: $P(h_j=1|v) = \text{sigmoid}(\Sigma\, w_{ij} \times v_i + c_j)$. Compute positive gradient: $\langle v_i \times h_j \rangle_{data}$

2. NEGATIVE PHASE (Fantasy): Reconstruct visible units from hidden: $P(v_i=1|h) = \text{sigmoid}(\Sigma\, w_{ij} \times h_j + b_i)$. Recompute hidden units from reconstruction. Compute negative gradient: $\langle v_i \times h_j \rangle_{model}$

3. WEIGHT UPDATE:

```
Δw_ij = learning_rate × (<v_i × h_j>_data - <v_i × h_j>_model)
Δb_i = learning_rate × (v_i_data - v_i_reconstructed)
Δc_j = learning_rate × (h_j_data - h_j_model)
```

4. Repeat for all training examples and epochs until convergence.

Applications of Boltzmann Machines:

- Dimensionality Reduction: Learning compact representations of data.
- Collaborative Filtering: Movie/product recommendation systems.
- Feature Learning: Automatically learn useful features from raw data.
- Image Recognition: Detect patterns in images.
- Natural Language Processing: Learn word and sentence representations.

SECTION 5: Explanation-Based Learning & Additional Topics

Exam Ref: 2079 Jestha Back, 2071 Magh, 2070 Magh, 2073 Bhadra

Q10. Define inductive learning. What is self-organizing Map (SOM)? Explain all the steps involved in SOM with suitable example. [2+5] (2074 Bhadra)

Answer: Inductive vs Explanation-Based Learning

Feature	Inductive Learning	Explanation-Based Learning (EBL)
Basis	Generalizes from multiple examples	Generalizes from a SINGLE example using domain knowledge
Data Required	Many training examples needed	One example + background knowledge sufficient
Speed	Slower — needs many examples	Faster — uses existing knowledge
Knowledge Used	Minimal prior knowledge	Rich domain knowledge required
Output	Statistical patterns/rules	Logical proof-based explanation
Example	Learn 'cup' from 100 images	Learn 'cup is liftable' from one example + physics laws

Self-Organizing Map (SOM):

A Self-Organizing Map (developed by Teuvo Kohonen, 1982) is an unsupervised neural network that produces a LOW-DIMENSIONAL (usually 2D) representation of high-dimensional input data, while preserving the topological structure of the data.

Also called: Kohonen Network or Kohonen Map.

Structure of SOM:

- Input Layer: n input neurons (one per feature of input vector).
- Output Layer (Map): A 2D grid of neurons (e.g., 5×5 = 25 neurons), each with a weight vector of the same dimension as input.
- No connections between output neurons — each output neuron competes to be the winner.

Steps in SOM Training Algorithm:

1. INITIALIZATION: Initialize all weight vectors w_j randomly (or using small random values).

2. SELECT INPUT: Present a training input vector x from the dataset.

3. FIND BEST MATCHING UNIT (BMU): Find the output neuron j* whose weight vector is CLOSEST to input x (minimum Euclidean distance).

```
j* = argmin_j ||x - w_j||
```

4. UPDATE WEIGHTS: Update the BMU and its NEIGHBORS in the grid to move closer to x.

```
w_j(t+1) = w_j(t) + α(t) × h(j*, j, t) × [x - w_j(t)]
```

Where: α(t) = learning rate (decreases with time), h(j*, j, t) = neighborhood function (Gaussian, decreases with distance and time).

5. DECREASE Learning Rate and Neighborhood Size over time (cooling schedule).

6. REPEAT: Go to Step 2 until the maximum number of iterations is reached or the map stabilizes.

Applications of SOM:

- Data visualization: Visualizing high-dimensional data in 2D.
- Document clustering: Grouping similar documents on a 2D map.
- Pattern recognition: Detecting patterns in sensor data.
- Market segmentation: Grouping customers by behavior.
- Anomaly detection: Identifying unusual patterns.

Q11. What is Natural Language Processing? Explain the different steps in Natural Language Processing (Note: This question overlaps with Chapter 7 but is frequently asked with Chapter 6). Explain fuzzy logic. [2+6] (Multiple years)

Answer: Fuzzy Logic — Additional Detail

Fuzzy Logic vs Classical Logic vs Probability:

Feature	Classical Logic	Probability	Fuzzy Logic
Truth Value	0 or 1 only	P between 0 and 1	Membership degree 0 to 1
Meaning	Absolutely true or false	Likelihood of being true	Degree of belonging to a set
Example	Age ≥ 18 is an adult (binary)	P(adult) = 0.7 (chance)	Adult membership = 0.7 (partial)
Used for	Crisp decision making	Statistical uncertainty	Vague/linguistic uncertainty

Membership Function Types:

- Triangular: Simple triangle shape — peaks at center value.
- Trapezoidal: Flat top — range of values with full membership.
- Gaussian: Bell curve shape — smooth, gradual transitions.
- Singleton: Sharp spike at exactly one value.

QUICK REVISION: Chapter 6 — Key Points Summary

Topic	Key Points to Remember	Expected Marks
Concepts of Learning	Definition, Types: Supervised, Unsupervised, Reinforcement — with examples	4-6
Learning by Analogy	4 Steps: Retrieve, Reuse, Revise, Retain (Case-Based Reasoning)	4-6
Inductive Learning / ID3	Entropy, Information Gain, Decision Tree construction step-by-step	6-8
Explanation-Based Learning	Single example + background knowledge → general rule	2-4
Genetic Algorithm	7 steps: Init → Fitness → Select → Crossover → Mutate → Replace → Terminate	6-8
Fuzzy Logic	Fuzzy sets, membership functions, 4-step inference: Fuzzify→Evaluate→Aggregate→Defuzzify	6-8
Boltzmann Machines	Energy function, visible+hidden units, RBM, Contrastive Divergence training	6-8
Self-Organizing Maps (SOM)	Unsupervised, 2D output grid, BMU, neighborhood update, topology preservation	4-6

Best of Luck! | TU IOE | BCT | CT 653 Artificial Intelligence | Chapter 6